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Course: Calculus Beta

Assignment: MyLab Math - Uge 8

Let A and B be independent events with $P(A) = \frac{1}{10}$ and $P(B) = \frac{1}{3}$. Find $P(A \cap B)$ and $P(A \cup B)$.

If A and B are independent events, $P(A \cap B) = P(A) \cdot P(B)$. Substitute each probability and multiply.

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{10} \cdot \frac{1}{3} = \frac{1}{30}$$

To find $P(A \cup B)$, use the union rule.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

$$\begin{aligned} P(A \cup B) &= \frac{1}{10} + \frac{1}{3} - \frac{1}{30} \\ &= \frac{3}{30} + \frac{10}{30} - \frac{1}{30} \\ &= \frac{12}{30} \end{aligned}$$

$P(A \cup B)$, in lowest terms, is $\frac{2}{5}$.